

Mateusz Grzonka

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SUMMARY

Backend-focused Computer Science student at Gdańsk University of Technology. Interested in music theory, game development, automotive engineering and tools that improve everyday life. Actively seeking a summer internship with a goal of establishing long-term professional collaboration

EDUCATION

Gdańsk University of Technology | B.Eng. in Computer Science **2023 – Expected 2027**

- Thesis: *A 3D FDTD-based acoustic wave propagation simulator with RL Agent improving room acoustics.*
- Publication: *"Assessment of the selected PML parameters on residual reflections in FDTD acoustic simulations"*

ZSŁ Technical Highschool in Gdańsk | IT Technician **2019 – 2023**

PROJECTS

A 3D FDTD-based Acoustic Wave Propagation Simulator *Python, CUDA, Taichi Lang*

- A high-performance 3D simulator utilizing the FDTD method to model acoustic wave behavior in complex environments. Consists of a Voxelizer, Wave Propagation Engine and Visualizer. Currently working on a RL Agent that helps you adapt your room for better general acoustics.

Secure TTP/Server/Client Authentication Protocol *C#, .Net*

- Designed and implemented a custom, three-node network protocol (Client, Server, TTP) with secure, encrypted communication channels using hybrid cryptography (RSA-4096 & AES-256).

Multiplayer Game with UDP Server *Python, Godot*

- Godot game and custom Python backend server to handle real-time multiplayer networking utilizing UDP protocols with custom frames to manage game state synchronization.

MACD Market Trading Strategy Calculator *Python*

- Developed an algorithmic calculator utilizing the MACD method to analyze market data. Implemented features to generate and visualize buy/sell signals based on simulated trading scenarios.

EXPERIENCE

Coding Giants | *Math and Programming Teacher* **2023 – 2025**

- Translated complex computer science concepts into easily understandable lessons. *Python*

Intel | *Hardware Specialist Internship* **2021**

- Collaborated with the engineering team to troubleshoot hardware integration issues. *Hardware*

SKILLS

- **Languages:** Python, C#, C/C++, Java, SQL, Taichi Lang, JavaScript, HTML/CSS
- **Frameworks & Architecture:** .NET, Spring, Microservices, RabbitMQ
- **Technologies & Tools:** Linux, Docker, Git, VirtualBox, Markdown
- **Soft Skills & Misc:** Technical Documentation, Research, Teaching, Teamwork
- **Spoken Languages:** English (C1), Polish (Native)

EVENTS

- Took part in hackathon "Hack4Change 2025" organized by CODE:ME.
- Took part in three different "Infoshare" Tech Events.